

INSTRUCTOR

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STUDENT

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ROLL NO:2K26/AI/55

PROGRAM:1

- Write a program that takes the user's name as input and then prints a greeting message like " Hello,Ali".

INPUT:

```
1  name = input("Enter your name: ")
2  print("Hello, " + name )
```

OUTPUT:

```
Enter your name: Maryam
Hello, Maryam
```

PROGRAM:2

- Write a Python program that takes hours worked and hourly rate as input from the user, then calculates and displays the gross pay.

INPUT:

```
1  hours = float(input("Enter hours worked: "))
2  rate = float(input("Enter hourly rate: "))
3
4  gross_pay = hours * rate
5
6  print("Gross Pay:", gross_pay)
```

OUTPUT:

```
Enter hours worked: 12
Enter hourly rate: 1000
Gross Pay: 12000.0
```

PROGRAM:3

- Write a Python program that takes temperature in Celsius as input from the user and converts it to Fahrenheit.

INPUT:

```
1 celsius = float(input("Enter temperature in Celsius: "))
2
3 fahrenheit = (celsius * 9/5) + 32
4
5 print("Temperature in Fahrenheit:", fahrenheit)
```

OUTPUT:

```
Enter temperature in Celsius: 37
Temperature in Fahrenheit: 98.6
PS C:\Users\Admin\py>
```

PROGRAM:4

- Write a Python program that takes two numbers as input and calculates the first number raised to the power of the second number (exponentiation).

INPUT:

```
1 num1 = float(input("Enter first number: "))
2 num2 = float(input("Enter second number: "))
3
4 result = num1 ** num2
5
6 print("Result:", result)
```

OUTPUT:

```
Enter first number: 55
Enter second number: 46
Result: 1.1394196811467816e+80
```

PROGRAM:5

What does the following Python code do? Explain the type conversion that happens in this program.

```
age=20

item=5

print("age",age)

print("items", item)

age = float(age)

item= float(item)

print("age:"age)

print("items", item)
```

INPUT:

```
1  age = 20
2  item = 5
3
4  print("age :", age)
5  print("items :", item)
6
7  age = float(age)
8  item = float(item)
9
10 print("age :", age)
11 print("items :", item)
```

EXPLANATION:

- Initially, age and item are integers (int), so their values are 20 and 5. The float () function converts these integer values into floating point numbers
- **20 → 20.0**
- **5 → 5.0**
- This process is called type conversion (Type casting)

OUTPUT:

```
age : 20  
items : 5  
age : 20.0  
items : 5.0
```